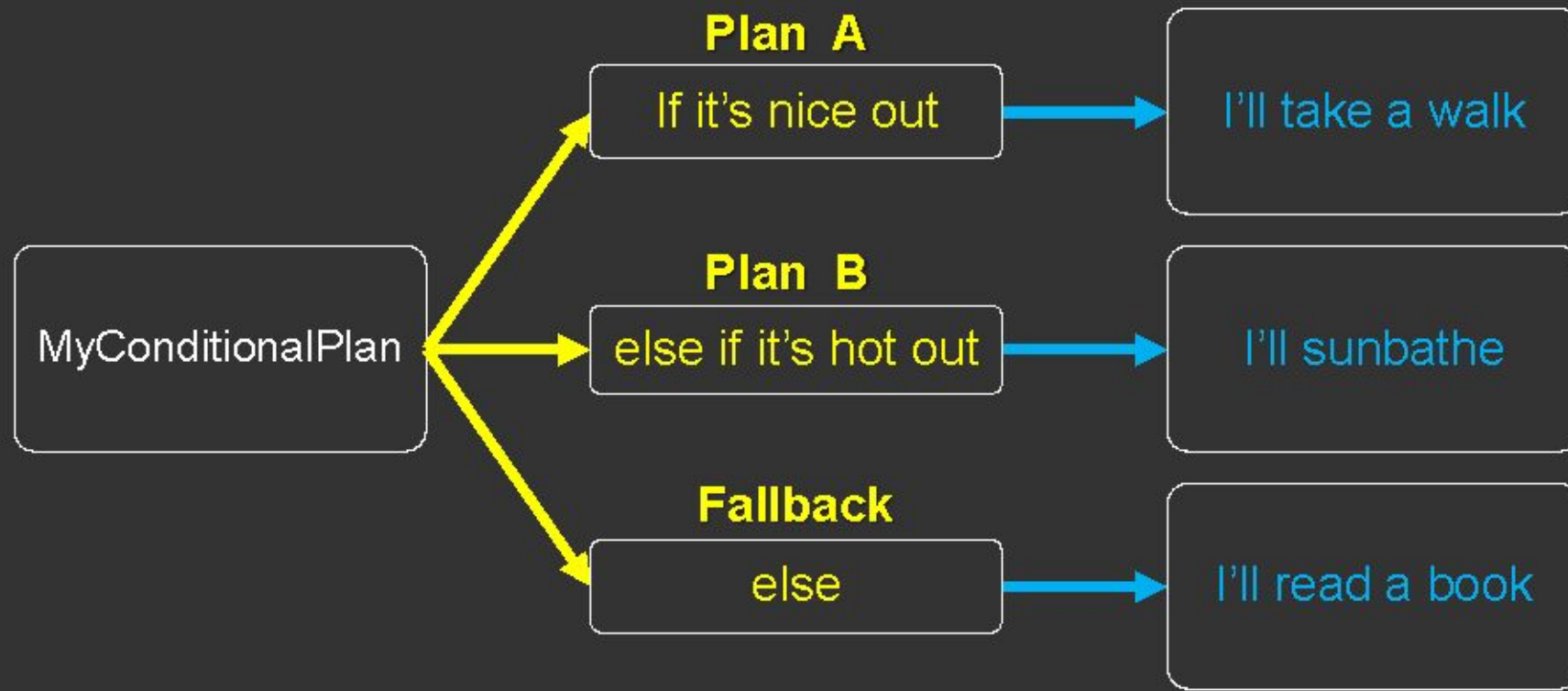


Conditionals

CS 141 – In Class Lecture

(slides by Jaime Spacco; edits by Vera Kazakova and David Bunde)

we may want to do things... conditionally



If statement (conditional)

```
if(condition)
{
    //body; only runs
    //if the condition is true
}
```

```
if(x < 10)
{
    System.out.println("Hello there");
}
```

Comparison operators

< less than
> greater than
== equal to
!= not equal to
<= less than or equal
>= greater than or equal

What does this print?

```
int num = 7;  
if(num < 0) {  
    System.out.print("positive number!");  
}  
System.out.print(" done ");
```

- A: positive number!
- B: positive number! done
- C: done
- D: (nothing)
- E: None of the above

What does this print?

```
int num = 7;  
if(num < 0) {  
    System.out.print("positive number! ");  
}  
System.out.print("done");
```

A: positive number!

B: positive number! done

C: done

D: (nothing)

E: None of the above

If statement (conditional)

```
if(condition) {  
    //body; only runs if the condition is true  
} else {  
    //only runs if the condition is false  
}
```

```
if(x < 10) {  
    System.out.println("Hello there");  
} else {  
    System.out.println("Goodbye");  
}
```

Nesting

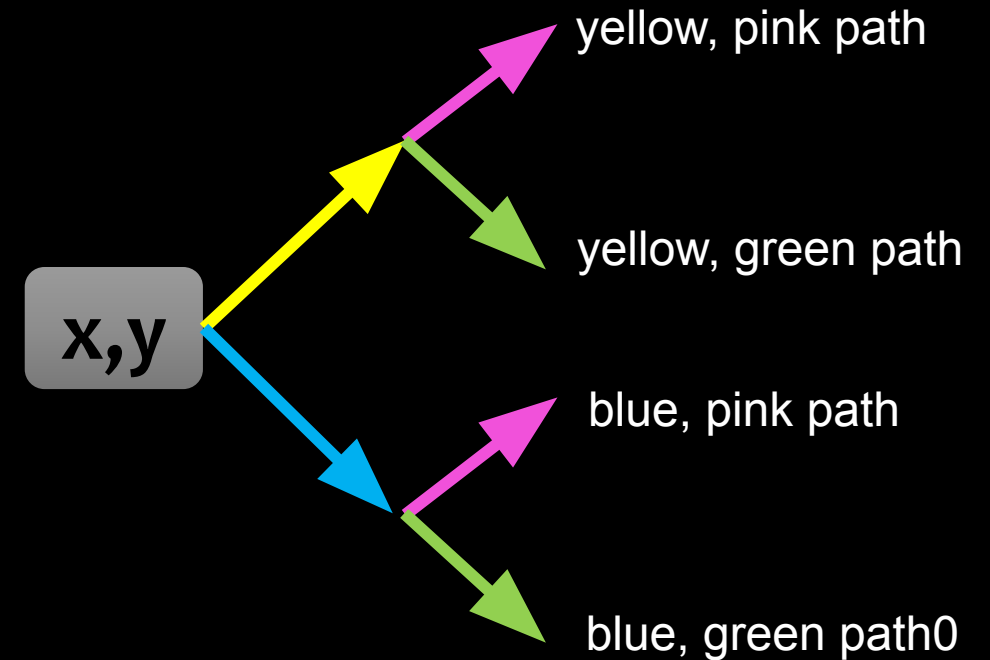
You can put if statements inside other if statements!

This lets you break your program down into a flow-chart

Nesting: statements inside other statements to create a flow-chart

//Note: assume we have some x and y (maybe as user input)

```
if (x<0) {  
    if (y<0) {  
        System.out.println("x<0 and y<0");  
    } else {  
        System.out.println("x<0 and y>=0");  
    }  
} else {  
    if (y<0) {  
        System.out.println("x>=0 and y<0");  
    } else {  
        System.out.println("x>=0 and y>=0");  
    }  
}
```



if / else if / else

```
if (CONDITION1) { //if condition is true
    ... //then do stuff
} else if (CONDITION2){//optional; only checked if CONDITION1=false
    ...
} else if (CONDITION3){//optional; only checked if CONDITION1 and CONDITION2 are false
    ...
}
... // as many "else if" as you want/need!
else {//optional; if none of the above... (SO CAN ONLY BE PLACED AT THE END!)
    ... // ...do this other "catch all" thing
}
```

```
int num=10;
if (num>0) {
    System.out.print("positive!");
}
else if (num<0){
    System.out.print("negative!");
}
else
    System.out.print("Zero!");
System.out.print(" ...done.");
```

Text adventure through campus with nested if-statements:

```
System.out.println("Go north (1) or south (2)?");
int choice1 = scan.nextInt(); // get an int from user (we'll show you how soon!)
if (choice1 == 1) {
    System.out.println("You are now in front of Old Main.");
    System.out.println("Enter Old Main (1) or head to GDH (2)?");
    int choice2 = scan.nextInt(); // get an int from user
    if (choice2 == 1) {
        // handle Old Main
    } else if (choice2 == 2) {
        // handle GDH
    } else {
        System.out.println("Illegal choice");
    }
} else if (choice1 == 2) {
    // Do whatever happens to the south
} else {
    System.out.println("Not a valid direction!");
}
```

What does this print when mode is assigned the value 1?

```
if (mode==1) {  
    System.out.print("mode is 1 ");  
} if (mode==2) {  
    System.out.print("mode is 2 ");  
} else {  
    System.out.print("not 1 or 2 ");  
}
```

A: mode is 1

B: mode is 2

C: mode is 1 mode is 2

D: mode is 2 not 1 or 2

E: mode is 1 not 1 or 2

What does this print when mode is assigned the value 1?

```
if (mode==1) {  
    System.out.print("mode is 1 ");  
} if (mode==2) {  
    System.out.print("mode is 2 ");  
} else {  
    System.out.print("not 1 or 2 ");  
}
```

A: mode is 1

B: mode is 2

C: mode is 1 mode is 2

D: mode is 2 not 1 or 2

E: mode is 1 not 1 or 2

**There is no ELSE
along with the 2nd if!**

**This is an example of
bad style, likely
leading to incorrect
functionality!**

What does this print when mode is assigned the value 1?

```
if (mode==1) {  
    System.out.print("mode is 1");  
}  
if (mode==2) {  
    System.out.print("mode is 2");  
} else {  
    System.out.print("not 1 or 2");  
}
```

There is no connecting “else if”, so we have two if-blocks. Clearer way would be to separate the two if-blocks:

```
if (mode==1) {  
    System.out.print("mode is 1");  
}  
  
if (mode==2) {  
    System.out.print("mode is 2");  
} else {  
    System.out.print("not 1 or 2");  
}
```

But that's probably not what you meant to do!

A: mode is 1

B: mode is 2

C: mode is 1 mode is 2

D: mode is 2 not 1 or 2

E: mode is 1 not 1 or 2

What does this print when mode is assigned the value 1?

```
if (mode==1) {  
    System.out.print("mode is 1");  
}  
if (mode==2) {  
    System.out.print("mode is 2");  
} else {  
    System.out.print("not 1 or 2");  
}
```

A: mode is 1

B: mode is 2

C: mode is 1 mode is 2

D: mode is 2 not 1 or 2

E: mode is 1 not 1 or 2

There is no connecting "else if", so we have two if-blocks. Clearer way would be to separate the two if-blocks:

```
if (mode==1) {  
    System.out.print("mode is 1");  
}  
  
if (mode==2) {  
    System.out.print("mode is 2");  
} else {  
    System.out.print("not 1 or 2");  
}
```

Probably meant to do a single if-block:

```
if (mode==1) {  
    System.out.print("mode is 1");  
}  
else if (mode==2) {  
    System.out.print("mode is 2");  
} else {  
    System.out.print("not 1 or 2");  
}
```

What gets printed here?

```
int x = 7;  
if (x < 10) {  
    System.out.print("Statement 1 ");  
} else if (x < 20) {  
    System.out.print("Statement 2 ");  
} else {  
    System.out.println("Statement 3 ");  
}
```

- A. Statement 1
- B. Statement 2
- C. Statement 1 Statement 2
- D. Statement 3
- E. It prints nothing.

What gets printed here?

```
int x = 7;  
if (x < 10) {  
    System.out.print("Statement 1 ");  
} else if (x < 20) {  
    System.out.print("Statement 2 ");  
} else {  
    System.out.println("Statement 3 ");  
}
```

- A. Statement 1
- B. Statement 2
- C. Statement 1 Statement 2
- D. Statement 3
- E. It prints nothing.

**Suppose this prints “statement 2”.
What must be true of num?**

```
if(num < 6) {  
    System.out.println("statement 1");  
} else {  
    System.out.println("statement 2");  
}
```

- A: greater than 6
- B: greater than or equal to 6
- C: less than 6
- D: less than or equal to 6
- E: not possible

**Suppose this prints “statement 2”.
What must be true of num?**

```
if(num < 6) {  
    System.out.println("statement 1");  
} else {  
    System.out.println("statement 2");  
}
```

A: greater than 6

B: greater than or equal to 6

C: less than 6

D: less than or equal to 6

E: not possible

What does this print when num is 6?

```
if(num < 6) {  
    System.out.print("less than 6 ");  
} else if(num > 6) {  
    System.out.print("greater than 6");  
}
```

A: less than 6 greater than 6

B: less than 6

C: greater than 6

D: it crashes

E: none of the above

What does this print when num is 6?

```
if(num < 6) {  
    System.out.print("less than 6 ");  
} else if(num > 6) {  
    System.out.print("greater than 6");  
}
```

A: less than 6 greater than 6

B: less than 6

C: greater than 6

D: it crashes

E: none of the above

This prints nothing at all!

Be careful! If you get no output from your program, then you may have a chained if that doesn't cover all conditions and you don't have an else to catch the remaining options

boolean values

Our conditions are actually expressions of a different primitive type: boolean

Every boolean has value either true or false

```
boolean myTruth = true;
```

Comparison operators take two numbers and give back a boolean:

```
4 < 3    ->  false
```

```
2 <= 10  ->  true
```

```
boolean isPos = x > 0;
```

AND, OR, NOT with booleans

A	B	A B	A && B	!A
F	F	F	F	T
F	T	T	F	T
T	F	T	F	F
T	T	T	T	F

how many of these are true?

```
boolean b1 = true && false;
```

```
boolean b2 = true || false;
```

```
boolean b3 = (true && false) || false;
```

```
boolean b4 = true || false && false;
```

A. 0

B. 1

C. 2

D. 3

E. 4

how many of these are true?

```
boolean b1 = true && false;
```

```
boolean b2 = true || false;
```

```
boolean b3 = (true && false) || false;
```

```
boolean b4 = true || false && false;
```

A. 0

B. 1

C. 2

D. 3

E. 4

ifs and booleans are not magical

`boolean` is just another primitive type with values `true` or `false`

Anywhere you are expecting a boolean value (such as in the `CONDITION` of an if statement), you can always put in a boolean *variable* or a *literal* or *something that evaluates to a boolean*.

Thus, these are equivalent:

```
int x=rand.nextInt();  
if (x > 0) {  
    System.out.println("positive");  
}
```

```
int x=rand.nextInt();  
boolean isPos = x > 0;  
if (isPos) {  
    System.out.println("positive");  
}
```

Similarly, these are the same

```
if (x > 0 && y <= z) {  
    // do whatever  
}
```

```
boolean b1 = x > 0;  
boolean b2 = y <= z;  
if (b1 && b2) {  
    // do whatever  
}
```

Note on comparator syntax

//You *cannot* do this:

```
if (10 < x < 20) { ← This will not even compile!
    // do something
}
```

//You must do this instead:

```
if (10 < x && x < 20) {
    // do something
}
```

Which inputs will print “yay!”?

```
int x = ____;  
int y = ____;  
if (x==0 || y==1) {  
    System.out.println("yay!");  
}
```

- A. 1
- B. 2
- C. 1,3,4
- D. 1,2,4
- E. 1,2,3,4

x	y	input #
0	0	1
0	1	2
1	0	3
1	1	4

Which inputs will print “yay!”?

```
int x = ____;  
int y = ____;  
if (x==0 || y==1) {  
    System.out.println("yay!");  
}
```

- A. 1
- B. 2
- C. 1,3,4
- D. 1,2,4**
- E. 1,2,3,4

x	y	input #
0	0	1
0	1	2
1	0	3
1	1	4

Which inputs will print “yay!”?

```
int x = ____;  
int y = ____;  
if (x==0 && y==1) {  
    System.out.println("yay!");  
}
```

- A. 1
- B. 2
- C. 1, 3, 4
- D. 1, 2, 4
- E. 1, 2, 3, 4

x	y	input #
0	0	1
0	1	2
1	0	3
1	1	4

Which inputs will print “yay!”?

```
int x = ____;  
int y = ____;  
if (x==0 && y==1) {  
    System.out.println("yay!");  
}
```

A. 1

B. 2

C. 1, 3, 4

D. 1, 2, 4

E. 1, 2, 3, 4

x	y	input #
0	0	1
0	1	2
1	0	3
1	1	4

DONE FOR NOW